Data Analytics (Experiment NO. 2)

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.datasets import load_iris

iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['species'] = iris.target
df['species'] = df['species'].map({0: 'Iris-setosa',
                                   1: 'Iris-versicolor',
                                   2: 'Iris-virginica'})
df.head()
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

Next steps: Generate code with df | New interactive sheet

```
for species in df['species'].unique():
    print("\nStatistical details for", species)
    print(df[df['species'] == species].describe())
```

```
min              4.30000          2.300000          1.000000
25%              4.80000          3.200000          1.400000
50%              5.00000          3.400000          1.500000
75%              5.20000          3.675000          1.575000
max              5.80000          4.400000          1.900000

       petal width (cm)
count         50.000000
mean           0.246000
std            0.105386
min            0.100000
25%            0.200000
50%            0.200000
75%            0.300000
max            0.600000

Statistical details for Iris-versicolor
```

```
50%             5.900000          2.800000          4.350000
75%             6.300000          3.000000          4.600000
max             7.000000          3.400000          5.100000

       petal width (cm)
count         50.000000
mean           1.326000
std            0.197753
min            1.000000
25%            1.200000
50%            1.300000
75%            1.500000
max            1.800000

Statistical details for Iris-virginica
       sepal length (cm)  sepal width (cm)  petal length (cm)  \
count           50.00000         50.000000          50.000000
mean             6.58800          2.974000           5.552000
std              0.63588          0.322497           0.551895
min              4.90000          2.200000           4.500000
25%              6.22500          2.800000           5.100000
50%              6.50000          3.000000           5.550000
75%              6.90000          3.175000           5.875000
max              7.90000          3.800000           6.900000

       petal width (cm)
count          50.00000
mean            2.02600
std             0.27465
min             1.40000
25%             1.80000
50%             2.00000
75%             2.30000
max             2.50000
```

```
grouped_stats = df.groupby('species').agg(['mean', 'median', 'min', 'max', 'stc
grouped_stats
```

	sepal length (cm)					sepal width (cm)					pe
	mean	median	min	max	std	mean	median	min	max	std	me
species											
Iris-setosa	5.006	5.0	4.3	5.8	0.352490	3.428	3.4	2.3	4.4	0.379064	1.4
Iris-versicolor	5.936	5.9	4.9	7.0	0.516171	2.770	2.8	2.0	3.4	0.313798	4.2
Iris-virginica	6.588	6.5	4.9	7.9	0.635880	2.974	3.0	2.2	3.8	0.322497	5.5

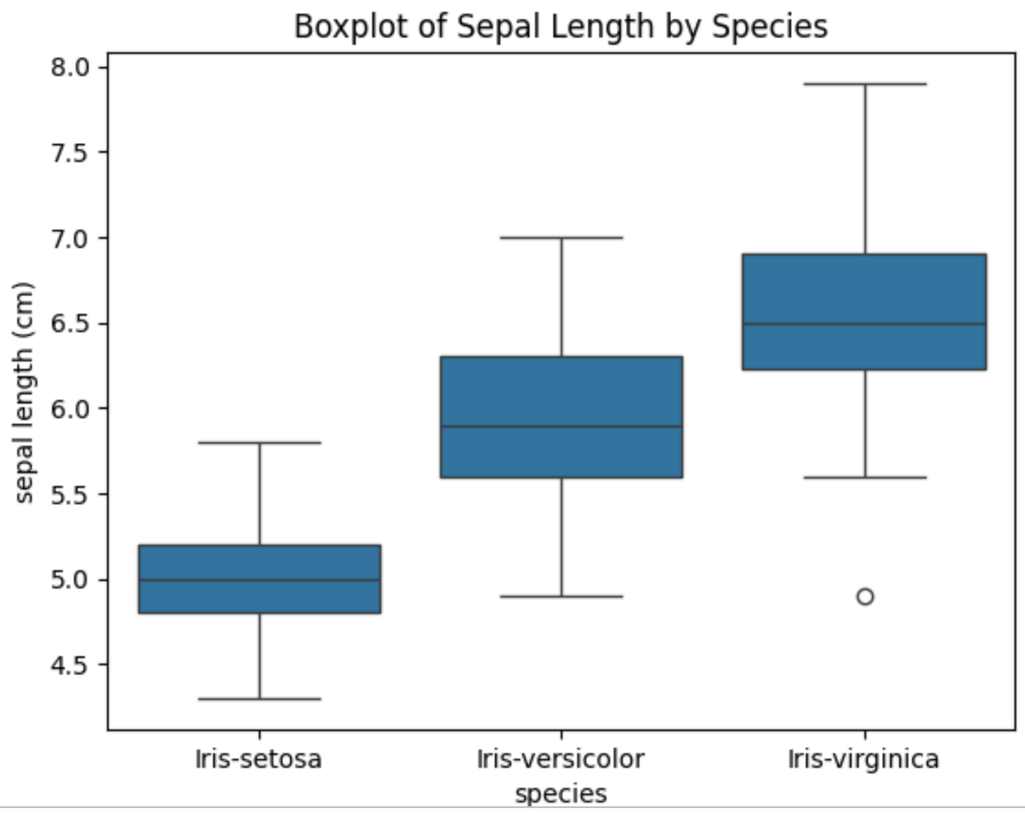
Next steps: Generate code with `grouped_stats` New interactive sheet

```
setosa = df[df['species'] == 'Iris-setosa']['sepal length (cm)'].tolist()
versicolor = df[df['species'] == 'Iris-versicolor']['sepal length (cm)'].tolist
virginica = df[df['species'] == 'Iris-virginica']['sepal length (cm)'].tolist()

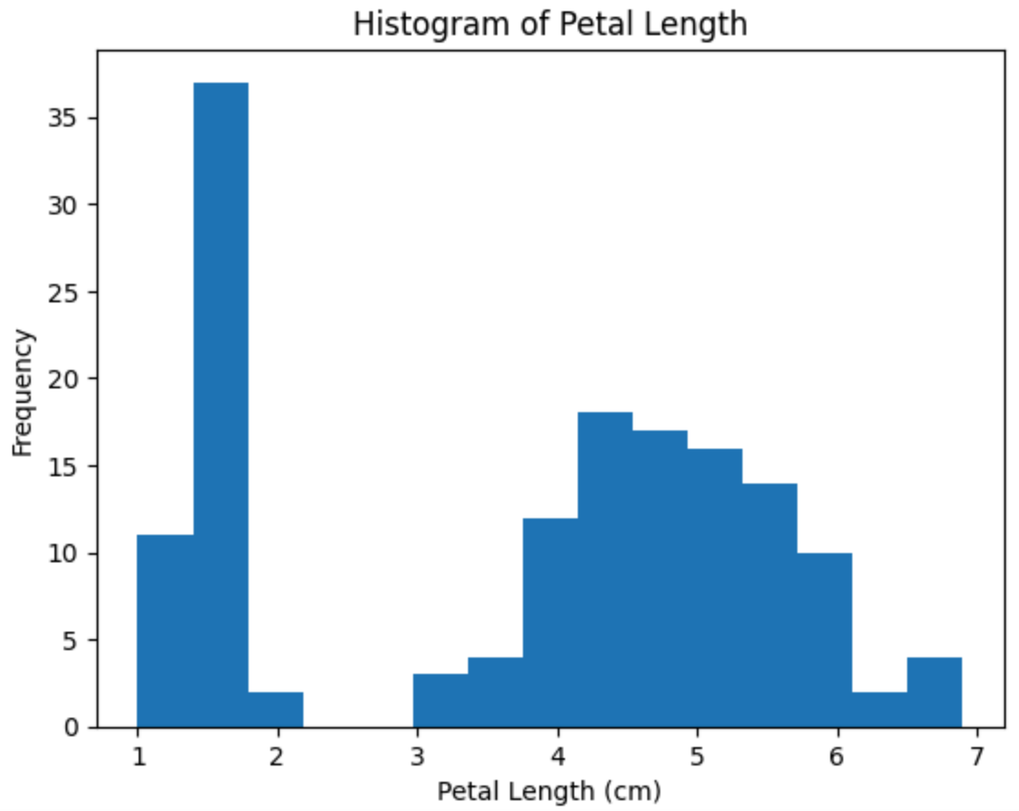
print("Setosa Sepal Length:", setosa[:5])
print("Versicolor Sepal Length:", versicolor[:5])
print("Virginica Sepal Length:", virginica[:5])
```

```
Setosa Sepal Length: [5.1, 4.9, 4.7, 4.6, 5.0]
Versicolor Sepal Length: [7.0, 6.4, 6.9, 5.5, 6.5]
Virginica Sepal Length: [6.3, 5.8, 7.1, 6.3, 6.5]
```

```
plt.figure()
sns.boxplot(x='species', y='sepal length (cm)', data=df)
plt.title('Boxplot of Sepal Length by Species')
plt.show()
```




```
plt.figure()
plt.hist(df['petal length (cm)'], bins=15)
plt.title('Histogram of Petal Length')
plt.xlabel('Petal Length (cm)')
plt.ylabel('Frequency')
plt.show()
```




```
plt.figure()
plt.scatter(df['sepal length (cm)'], df['petal length (cm)'])
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Petal Length (cm)')
```

```
plt.title('Scatter Plot: Sepal Length vs Petal Length')
plt.show()
```

